Ayan Bin Saif

ayan.binsaif@uwaterloo.ca | linkedin.com/in/stitches | github.com/draggle | ayans.dev

Education

University of Waterloo — Waterloo, ON
Honours Bachelor of Mathematics (Co-op) — Present
Applied Mathematics with Scientific Computing and Scientific Machine Learning

Work Experience

TERN — May 2026 — Aug 2026
Engineering Intern — Austin, Texas, United States

- Engineered a GPS-free cold-start localization stack using scored road-graph matching and persisted position priors, delivering 5.2x more position fixes at a 2.6x lower error rate, validated on a 251-drive RTK corpus.
- Rebuilt the offline road-graph builder (OpenStreetMap → matching graphs), eliminating non-determinism across runs and threads, while achieving 3.8x faster runtime and 53% lower peak memory.
- Built the RTK ground-truth evaluation harness used as the ship gate for localization changes, with per-drive scoring that uncovered a bug understating baseline accuracy by 5x.
- Built a magnetometer validation channel that learned per-vehicle compass calibration and vetoed incorrect heading candidates, catching localization failures up to 29 km away that other signals could not detect.

Apple — Feb 2024 — Jul 2024
iOS App Developer (Mentorship) — Toronto, Ontario, Canada

- Built native iOS features in Swift and SwiftUI as part of a selective technical Apple mentorship in iOS development, emphasizing performance optimization and scalable UI architecture.
- Architected EduBuddy, an iOS productivity app with a personalized recommendation engine; increased beta engagement by 30% through HIG-aligned UX design and performance optimization.
- Reduced UI load times by 400ms and memory usage by 25% via lazy loading, view hierarchy refactoring, and SwiftUI lifecycle optimization.
- Owned full SDLC from wireframing to TestFlight deployment; presented a technical demo to Apple iOS engineers and was selected for showcase among 200+ participants.

Projects

AlphaHedge | Next.js, TypeScript, Supabase, Google Gemini, Sim.ai — Jan 2026

- Developed an AI-driven investment research platform generating structured multi-factor stock analysis reports in under 2 minutes, built at the Y Combinator Full-Stack Hackathon.
- Engineered a multi-agent council with 4 autonomous agents orchestrated via Sim.ai to independently analyze stocks and converge on ratings via weighted decision aggregation and a contextual RAG pipeline.
- Built an asynchronous job-polling architecture for multi-agent workflows; engineered a UI that parses raw LLM outputs into styled badges and interactive financial tooltips.

Rate My Rez | React, Firebase, Tailwind CSS — Nov 2025

- Engineered a full-stack housing review platform for University of Waterloo students using React and Firebase; reached 10,000+ unique visits in under 1 month of launch.
- Reduced UI state recomputation by 60% through React memoization; maintained sub-100ms update latency via Firestore synchronization.

Technical Skills

Languages:	Python, C, C++, TypeScript, JavaScript, Swift, Bash, Java, SQL, HTML/CSS
Frameworks:	Next.js, React, React Native, SwiftUI, Node.js, Tailwind CSS, OpenCV, MediaPipe
Tools:	Docker, Supabase, PostgreSQL, Firebase, Google Gemini, Git, Unix/Linux, Vercel, AWS
Core Concepts:	Multi-Agent Systems, RESTful API, OOP, Data Structures & Algorithms